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SCENARIO BASED

CSA1721- CLOUD COMPUTING & BIG DATA ANALYTICS

Q.1) A media company wants to migrate its Video streaming platform to the cloud.

To support million of concurrent user with minimal buffering, the media company should adopt a cloud-based scalable architecture.

① Cloud Load Balancer

→ A Load balancer receives all user request.

→ It distribute traffic equally among multiple application server

② Application server

- Multiple Virtual machine or containers host the streaming application.

③ Object storage

- Videos are stored cloud object storage such as Amazon S3, Azure Blob storage or Google cloud storage

Content delivery network (CDN)

→ CDN stores cached copies of Videos in edge location close to the users.

Load Balancing and Auto scaling

Load Balancer

→ Distribute incoming request evenly across all servers

→ Removes unhealthy servers from service

Auto Scaling

→ Automatically increase the number of server during peak hours

→ Removes extra server when traffic decreases.

Cost optimization

→ Pay as you Go pricing

→ use reserved instance for predictable workload.

→ Move old Videos to low cost archival storage

→ Use CDN caching to reduce bandwidth cost

→ Monitor resource usage and optimize server size.

2) A fintech startup is deploying application using container across multiple cloud providers

A fintech company requires consist application deployment across different cloud platform while ensuring high avaibility and reliability

Container archestration

The best solution is to use Kubernetes

Kubernetes

- Deploys and manages containers unconditionally
- Restart failed container
- Perform rolling update with no downtime
- Balance workload among nodes

Multi cloud strategy

Application are deployed on multiple cloud providers such as AWS, Microsoft Azure, and Google cloud platform

→ Better avaibulity
- Disaster recovery
- No dependency

Service Discovery

Kubernetes automatically assigns DNS names to service.
when one container needs another service.

→ It communicate using the service name
→ Kubernetes automatically redirect traffic to healthy container

Scaling

Horizontal Pod Autoscaler monitor CPU and memory
when demand increases
- More container are created

when demand increases
- Extra container are removed.

- High availability
- Vendor independence

- More complex management
- Higher network cost

Q.3) An e learning platform experience data loss due to accidental deletion

To prevent data loss, the organization should implement a strong cloud native backup and disaster recovery plan

Backup strategy
- schedule automatic backup every day
- Enable object Versioning so deleted files can be restored
- Replicate backup to another geographical region.

Disaster Recovery

If the primary region fails.

→ Backup server become active
→ Data is restored from replicated copies
→ Application resume with minimal downtime

Recovery Time objective (RTO)

RTO is the maximum acceptable time required to restore service

→ Automatic failover
→ standby server
→ fast recovery procedure

Recovery point objective (RPO)

RPO is the maximum amount of data loss acceptable

- continuous backup
- Real-time replication

Business continuity steps

- Maintain backup copies in multiple region.
- Test disaster recovery plans regularly
- Enable automatic failover
- Monitor backup health.
- Train employee on recovery procedure.

Q4) Government agency adopting a hybrid cloud model.

Government agencies often keep sensitive information on premises while moving public services to the cloud.

Hybrid cloud architecture

- Sensitive citizen data remain in the on-premises data centre
- public application are hosted in public cloud.
- A secure VPN or dedicated private connection connects both the environment

Identity Management

Security Measures Include

- Multifactor Authentication (MFA)
- Role based Access control (RBAC)
- Least priviledge access
- Single Sign in (SSO)

Encryption

Data should be encrypted

→ At rest using AES-256

- In transit using SSL/TLS protocols

→ Managing two infrastructure

→ synchronizing data

→ Higher operational cost

→ compliance with government regulation

→ Complex monitoring and maintenance.

Q.5) SaaS Provider requires high availability and fault tolerance

To ensure uninterrupted service the SaaS application should be deployed across multiple cloud region.

Multi region deployment

- Deploy identical application server in two or more cloud region

- use a Global Load Balancer to direct user to the nearest healthy region

- If one region fails user are redirected automatically to another region.

Data Replication

Two methods

Synchronous Replication - data is copied immediately (high consistency)

Asynchronous Replication
Data is copied after a short delay (better performance

Failover Mechanism

- Health check detect the failure
- Traffic is redirected automatically
- Secondary database become active

Monitoring and Alerting

- CPU usage
- Memory usage
- Storage
- Network traffic

Alerts are sent through

- Email
- SMS
- Mobile notification